Free Resolutions

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This is a note discussing the basic construction of Free Resolutions and the computation of free resolutions for various modules.

1 Basic Construction

Let R be a commutative ring with 1 and let M be a finitely generated module over R. If M is generated by $x_1, \cdots, x_n$ elements with m relations f_i between the generators then recall that we can encode the presentation of M with an exact sequence

$$R^m \to R^n \to M \to 0$$

where $R^n \to M$ is given by $a_i \to x_i$ and where $R^m \to R^n$ is given by $b_i \to f_i(a_i)$. This is already pretty good. However, we can go further. Recalling that R-mod is an abelian category gives us that $\ker(R^m \to R^n)$ is also an R-module. And so we can ask for a presentation of $\ker(R^n \to M)$. This gives us a further exact sequence

$$R^{m_2} \to R^{n_2} \to R^m \to R^n \to M \to 0.$$

Then, we are able to iterate this procedure until we no longer have any relations between the set of generators of the current iterate. This process may not terminate (even when M is fg? we should find an example). Repeating this procedure gives the *free resolution of M*

$$\cdots \to R^{k_n} \to \cdots R^{k_2} \to R^{k_1} \to R^{k_0} \to M \to 0.$$

This is called "Free" since every object in the chain (except the final one) is a free module. Below we will compute some examples of free resolutions of various modules. [1]

[1] I am curious to note some basic thoery here. I suppose it is clear that a fg module should always have a free resolution. Is it clear that a given module has a unique free resolution (I suppose the above procedure can only spit

2 Examples

These examples are sorted into an order of increasing complexity. However, in all the following examples, R is a polynomial ring and M is a quotient of R. One can check these examples in Macaulay2 with the code what's the use of verbatim

Example 1. Let $R = k[x]$ and $M = R/(x^n)$ for some fixed n. Then notice that M is generated by $\bar{1} \in R/(x^n)$ as an R-module. Moreover, we only have a single relation $x^n \cdot \bar{1} = 0$. We can then capture this in the presentation

$$R^1 \xrightarrow{1 \mapsto x^n} R^1 \xrightarrow{1 \mapsto \bar{1}} M \to 0.$$

The kernel of $R^1 \to M$ is exactly the ideal (x^n) (as expected). Interpreting (x^n) as an R-module we find that it is a rank 1 free module (generated by x^n). That is a presentation for (x^n) as an R-module is

$$0 \to R^1 \xrightarrow{1 \mapsto x^n} (x^n).$$

Is the above sequence correct? Now since $\ker(R^1 \to (x^n)) = 0$ we can no longer iterate our procedure. This leaves us with a (the?) free resolution

$$R^1 \xrightarrow{1 \mapsto x^n} R^1 \xrightarrow{1 \mapsto x^n} R^1 \xrightarrow{1 \mapsto \bar{1}} M \to 0.$$

The final map does not give us any additional information (what do I mean by this? It's clear that we should trim the sequence however) and so we actually call the following sequence the free resolution of $M = R/(x^n)$.

$$R^1 \xrightarrow{1 \mapsto x^n} R^1 \xrightarrow{1 \mapsto \bar{1}} M \to 0.$$

This example shows us that whenever our kernel module is free we are done.

Example 2. Let $R = k[x, y]$ and $M = R/(x^2, y^2)$ elements in M are of the form $f = a_0 + a_1 x + a_2 y + a_3 xy$. As an R-module M is generated by $\bar{1} \in M$. We have the relations $x^2 = 0$ and $y^2 = 0$. We need to describe what it means for these relations to span all the kernel relations A presentation for M is thus

$$R^2 \xrightarrow{\begin{bmatrix} x^2 & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \to \bar{1}} M \to 0$$

Now, the kernel module $\ker(R^1 \to M) = (x^2, y^2)$, interpreting the ideal as an R-module. This module is generated by two elements x^2, y^2 and we have the R-linear relation that $y^2(x^2) - x^2(y^2) = 0$. Thus

out one exact sequence, but if we define a free resolution to be an exact sequence whose tail is the presentation then is there only one such sequence?)

our next iterated sequence is

$$R^1 \xrightarrow{\begin{bmatrix} y^2 \\ -x^2 \end{bmatrix}} R^2 \xrightarrow{\begin{bmatrix} x^2 & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \to \bar{1}} M \to 0.$$

Moreover the kernel module $\ker(R^2 \to (x^2, y^2)) = 0$ and so we are done. The above sequence is the free resolution for $M = R/(x^2, y^2)$.

Example 3.